

Lecture 26: Stochastic Processes Part 1

Strict Sense Stationary (SSS), Wide Sense Stationary (WSS), and Ergodic Processes

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1 Introduction to Stochastic Processes

A **stochastic process** $X(\zeta, t)$ is defined as an entire collection of waveforms or functions over time that are dependent on the outcome of a random experiment[cite: 27, 34].

1.1 Key Concepts

- **Realization:** A specific measured waveform from an experiment is called a realization of the process[cite: 103, 109].
- **Random Variable (R.V.):** If the process is "frozen" at a specific time $t = t_1$, the output evaluates to a single random variable, denoted as $X(\zeta, t_1) = X(t_1)$ [cite: 110, 111].
- **Multiple Variables:** By evaluating the process at multiple time instances $(t_1, t_2, \dots, t_n)$, we acquire multiple dependent random variables[cite: 120].

2 Process Characterization

Stochastic processes are characterized by their Probability Density Functions (PDF)[cite: 126, 131]:

- **1st Order:** Describes the R.V. at a single time t_1 : $f_{X_1}(x, t_1)$ [cite: 132, 133].
- **2nd Order:** Describes the joint probability across two times t_1 and t_2 : $f_{X_1 X_2}(x_1, x_2, t_1, t_2)$ [cite: 141, 142].
- **n^{th} Order:** Provides a complete statistical description: $f_{X_1 \dots X_n}(x_1, \dots, x_n, t_1, \dots, t_n)$ [cite: 152, 153].

3 Statistical Moments

3.1 Mean Function

The mean is the expected value of the random variable at a given time t [cite: 160].

$$\mu_X(t) = E[X(t)] = \int_{-\infty}^{\infty} x \cdot f_X(x, t) dx \quad (1)$$

Generally, the mean changes as a function of time[cite: 172].

3.2 Autocorrelation Function

Measures the relationship between the process sampled at two different times t_1 and t_2 [cite: 186].

$$R_{XX}(t_1, t_2) = E[X(t_1)X^*(t_2)] \quad (2)$$

Properties:

- **Conjugate Symmetry:** $R_{XX}(t_1, t_2) = R_{XX}^*(t_2, t_1)$ [cite: 210, 237].
- **Positive Definiteness:** The function forms a positive definite matrix[cite: 241]. For any complex constants a_i : $\sum_{i=1}^n \sum_{j=1}^n a_i a_j^* R_{XX}(t_i, t_j) \geq 0$ [cite: 277].

3.3 Autocovariance Function

Measures the relationship between the deviations of the process from its mean[cite: 253].

$$C_{XX}(t_1, t_2) = E[(X(t_1) - \mu_X(t_1))(X^*(t_2) - \mu_X^*(t_2))] \quad (3)$$

It is related to autocorrelation by: $C_{XX}(t_1, t_2) = R_{XX}(t_1, t_2) - \mu_X(t_1)\mu_X(t_2)$ [cite: 655].

4 Stationary Stochastic Processes

4.1 Strict Sense Stationary (SSS)

A process is SSS if its entire joint distribution is invariant under a time shift τ [cite: 856].

$$f_X(x_1, \dots, x_n; t_1, \dots, t_n) = f_X(x_1, \dots, x_n; t_1 + \tau, \dots, t_n + \tau) \quad (4)$$

For an SSS process, the mean μ_X is constant and the autocorrelation $R_{XX}(t_1, t_2)$ depends only on the time difference $t_1 - t_2$ [cite: 459, 478].

4.2 Wide Sense Stationary (WSS)

WSS is a weaker condition often used in practice[cite: 483]. A process is WSS if:

1. **Constant Mean:** $E[X(t)] = \mu_X = \text{constant}$ [cite: 492, 500].
2. **Lag-Dependent Autocorrelation:** $E[X(t_1)X^*(t_2)] = R_{XX}(\tau)$, where $\tau = t_1 - t_2$ [cite: 501, 502].

Note: SSS implies WSS, but WSS does not necessarily imply SSS unless the process is Gaussian[cite: 510, 523].

5 Ergodic Process

A process is **ergodic** if its time averages are equal to its ensemble averages[cite: 680, 688].

- **Time Average (Mean):** $\langle X(t) \rangle = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt$ [cite: 719, 725].
- **Time-Average Autocorrelation:** $R_{XX}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t)x(t+\tau) dt$ [cite: 727, 728].

In an ergodic process, a single realization can represent the entire process[cite: 708].

6 Summary of Mathematical Tools

Function	Formula
Mean Function	$\mu_X(t) = E[X(t)]$ [cite: 635]
Autocorrelation	$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)]$ [cite: 646]
Autocovariance	$C_{XX}(t_1, t_2) = R_{XX}(t_1, t_2) - \mu_X(t_1)\mu_X(t_2)$ [cite: 655]
Cross-Correlation	$R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)]$ [cite: 666]